

Toward Scalable Bayesian Uncertainty for Machine Learning with the Ray Tracing Sampler

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Abstract

This third report moves the study from MNIST classification to a real scientific target: regressing the alpha-element abundance $[\alpha/M]$ of 126,156 stars from their Gaia XP spectral coefficients, the dataset behind Laroche & Speagle (2025), with real, per-star label errors. The headline lesson is that the sampler was never the bottleneck, the noise model was: with the survey errors taken at face value the sampled posterior is overconfident by a factor of seven (z standard deviation 6.88 on held-out stars), a global intrinsic-scatter parameter repairs the average (0.95) but not the structure (0.39 to 1.43 across label-error bins), and a per-star noise channel $\sigma(x)$ makes calibration flat (1.08, bins 0.97 to 1.07). A Student-t likelihood with sampled degrees of freedom ($\nu = 5.48$) then fixes the tail shape, cutting the stars past the $|z| = 4$ tail from 108 to 14 against a chance rate of 1.6, and posts the best test log-likelihood of every method we ran. Sampling still matters: on the identical likelihood, an Adam MAP fit fails per-bin calibration outright (1.26 down to 1.04, 209 tail stars), a 10-seed deep ensemble inverts the staircase by over-padding precise stars, and tuned SGHMC runs measurably hot (kinetic temperature 1.08 to 1.09) with worse structure; only the ray tracing chain is flat, and it holds temperature by construction rather than by tuning a friction against the gradient-noise scale. We back the converged chain with an exactness certificate: a full-batch Metropolis finisher accepts its endpoint at 0.98, at the cost of a 10x smaller step, which is the measured version of the scaling argument for minibatch gates. Total campaign cost: roughly 90 dollars of cloud GPU time.

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🔗 **Code:** <https://github.com/YasirAlsugair/ray-tracing-sampler-experiments>

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1 Introduction

Reports 1 and 2 built and settled a controlled testbed: small MNIST classifiers whose posterior is exact by construction, on which the paper’s noise-softened accept test (Eq. 33 of [1]) delivered the study’s only formally converged chains, and the converged posterior’s uncertainty proved genuinely useful. The obvious objection to everything so far is that MNIST is a toy. This report answers it by moving the whole harness to a real scientific problem with real, heteroscedastic measurement errors: predicting the alpha-element abundance $[\alpha/M]$ of stars from their Gaia XP spectrophotometric coefficients (55 BP plus 55 RP per star), using the dataset behind [2], whose labels come with a quoted per-star uncertainty.

The move changes the question. On MNIST the ask was whether the sampler converges and whether its uncertainty beats baselines. On a survey regression the ask is the one astronomers actually care about: are the error bars honest, star by star? That splits into three questions this report works through. First, what does it take for the sampled posterior’s uncertainties to be calibrated on held-out stars, and where does miscalibration come from when it appears? Second, is good calibration a property of the likelihood (so any fit would inherit it) or of the sampling (so optimizers and cheap approximations lose it)? Third, how does ray tracing compare, on the same target and budget, against the standard stochastic-gradient MCMC baseline, SGHMC [3], whose known failure mode (gradient-noise heating) is exactly what the ray tracing construction is designed to remove?

2 Methodology

Data. We reproduce the pristine sample selection of [2] from the released catalog, landing at 126,156 stars against the paper’s 123,804 (within 1.9%; the residual difference traces to this table’s error columns not being exactly the astroNN ones). Inputs are the 110 XP coefficients, standardized on the training split; targets are APOGEE $[\alpha/M]$ labels with their per-star errors. The split is 80/20 (100,924 train, 25,232 held out) at a fixed seed. All predictive and calibration numbers in this report are on the held-out stars, and chain uncertainties use the full mixture predictive (label error plus mean member variance plus member disagreement).

Model and priors. The network is an MLP (110-64-64-1, tanh), $D = 11,329$ parameters, growing to 11,395 as the noise model gains parameters. The prior is Glorot-style per layer [4] (bias $\sigma = 0.1$); a matched $\mathcal{N}(0, 1)$ run serves as the prior ablation.

The noise-model ladder. We treat the likelihood itself as the experimental variable, in four nested rungs: (0) label errors taken at face value; (1) one global intrinsic-scatter parameter s , sampled with the weights; (2) a heteroscedastic per-star channel, $\sigma(x) = \exp(-3 + r(x))$ from a second network output, giving $\text{var}_i = \text{err}_i^2 + \sigma(x_i)^2$; (3) a Student-t likelihood with scale $\sigma(x)$ and sampled degrees of freedom ν . Each rung was run to convergence with the minibatch ray tracer and Eq. 33 gate, as in Report 2.

Convergence and exactness protocol. A chain is declared converged only when every monitored series (data misfit, weighted norm, noise-model parameters) is level under a drift-versus-noise rule, always read together with the acceptance rate, since a frozen all-rejecting chain passes any stationarity test. The converged endpoint is then stamped by an exact finisher: full-batch gradients with the exact Metropolis test, no minibatch approximation, launched from the endpoint. We also measure the Eq. 33 gate’s noise ceiling on this target by tuning ladders from fitted states.

Baselines, same likelihood. To separate sampler from likelihood, three baselines fit the identical rung-2 target with the identical warm-start schedule: an Adam MAP fit (three learning

rates, best training loss kept), a 10-seed deep ensemble [5] with the total-variance predictive, and SGHMC [3] in the physical parameterization, validated at $T = 1$ on an exact-gradient Gaussian, with a per-step kinetic thermometer $\langle v^2 \rangle$ (stationary value 1) so heating is read directly rather than inferred. All sampling ran on rented cloud GPUs; the full campaign, tuning grids included, cost roughly \$85 to \$90.

3 Results

3.1 The prior decides what the budget buys

Report 2 ended with burn-in as the dominant cost. On this target the cost disappeared, not because of the sampler but because of the prior. The Glorot prior is born on its own typical shell (weighted norm 11,235 against $D = 11,329$), so the chain spends its entire budget fitting the data and reaches RMSE 0.0492, a third better than a k -NN baseline (0.0742) on the same inputs. The $\mathcal{N}(0, 1)$ ablation, with the identical sampler and budget, stays pinned to its own shell, ends at 2.9 $\times$ the data misfit and RMSE 0.0704, barely at k -NN level. The 20,000-trajectory walk that dominated the MNIST study was a prior problem, not a sampler problem.

3.2 The noise-model ladder fixes calibration

Taking the survey's label errors at face value, the sampled posterior is spectacularly overconfident on held-out stars: the standardized residuals have a standard deviation of 6.88 where a calibrated model gives 1 (Figure 1). Member disagreement barely dents this (6.23). The repair is a ladder of likelihoods, not a different sampler. One global intrinsic-scatter parameter s , sampled jointly with the weights, lands at $s = 0.0452 \pm 0.0010$ and calibrates the average (z std 0.95) while failing the structure: across four label-error bins the z std runs 0.39 to 1.43, so precise stars get padded and noisy stars stay overconfident. Promoting the scatter to a per-star network output $\sigma(x)$ makes the structure flat: bins 0.97 / 1.07 / 1.07 / 1.07 at overall SD(z) 1.08 (Table 1). The lesson of the week in one line: the sampler was never the bottleneck, the noise model was.

3.3 Convergence, the gate's noise ceiling, and the exact stamp

The Eq. 33 gate turned out to be a diagnostic here, not a filter. At a prior draw its noise scale is astronomical (26M nats for Glorot, 3.5B for $\mathcal{N}(0, 1)$), so it accepts everything and must be tuned from a fitted state. At the fit it correctly all-rejects overly large steps (the luminosity ledger claims about 10M spurious nats per window at $dt = 10^{-4}$), and tuned by the paper's recipe its acceptance plateaus at a measurable noise ceiling (0.38 on the first likelihood, 0.45 on the corrected one): at the plateau its decisions are coin flips. The evidence the dynamics were nonetheless clean is threefold: an ungated control reproduces the gated chain's predictions to 1% (RMSE 0.0486 vs 0.0492), the drift rule with healthy acceptance declares every monitored series level, and the exact finisher accepts the endpoint (below).

Convergence itself was a campaign. The heteroscedastic chain's weighted norm marched for nine 500k-step legs without decelerating; escalating the batch from 1,024 to 4,096 at the same step size halved the gate's noise scale, and the march did not slow at first, which ruled out gradient-noise heating: the chain was genuinely still descending, and with cleaner gradients it descended faster, then decelerated, plateaued, and reversed. All three series went level at 16 million steps. Convergence also had an honest price: the in-transit global- s chain posts the campaign's best RMSE (0.0466), and the converged posterior bulk relaxes to 0.0494, the intrinsic residual floor. Numbers read off an in-transit chain flatter, which is the same warning Report 2 issued for the CNN, now on real data.

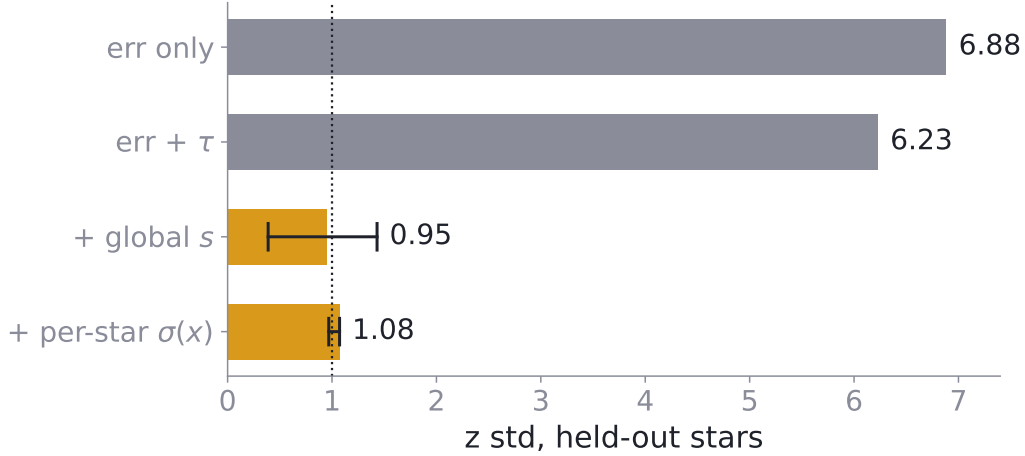


Figure 1 Calibration of the sampled posterior on 25,232 held-out stars as the likelihood climbs the noise-model ladder. A calibrated model has z std 1 (dotted). Survey errors alone leave the posterior overconfident by a factor of seven; a global scatter s fixes the average; the per-star $\sigma(x)$ fixes it while also being flat across label-error bins. Whiskers span the four label-error bins.

Table 1 The noise-model ladder, all rungs sampled to convergence with the minibatch ray tracer on the same data. Bin range is the z std across the four label-error bins; tails count held-out stars past the $|z| = 4$ equivalent (chance rate 1.6). The Student-t tail count uses the fitted t 's own tail probability.

likelihood	D	steps	RMSE	z std (bin range)	tails
survey errors only	11,329	2M	0.0492	6.88	
+ global scatter s	11,330	2M	0.0466	0.95 (0.39 to 1.43)	41
+ per-star $\sigma(x)$	11,394	16M	0.0494	1.08 (0.97 to 1.07)	108
+ Student-t, sampled ν	11,395	8.5M	0.0480	1.12 (1.05 to 1.14)	14

The exactness certificate: a full-batch Metropolis finisher launched from the converged endpoint accepts at 0.98 and reproduces its predictions (RMSE 0.0469), so the converged state is not a minibatch artifact. The cost is the point: the exact test rejects everything down to a step 10x smaller than the production minibatch step (a cliff, not a slope, in the pilot ladder). That is the scaling argument for minibatch gates, measured on our own target at $D = 11,330$.

3.4 Same likelihood, four fits: structure is a sampling property

Is flat calibration just a property of the corrected likelihood? No. Three baselines fit the identical rung-2 target with the identical warm-start schedule (Figure 2, Table 2). The Adam MAP fit fails the structure test outright: a staircase from 1.26 down to 1.04 across the bins and double the tail stars, and across its learning-rate sweep calibration *anti-correlates* with training loss, so descending deeper into the mode buys accuracy by spending calibration. The 10-seed deep ensemble is a genuine competitor on aggregates (best RMSE, best tails) but inverts the staircase: its member spread over-pads the most precise stars by 20%. Tuned SGHMC lands hot everywhere (next subsection). The cleanest summary is the maximum bin deviation from 1: chain 0.07, ensemble 0.21, MAP 0.26. Only the sampled posterior distributes uncertainty correctly star by star, and on the predictive scoreboard (Figure 3) the two ray tracing chains post the two best test log-likelihoods, with the Student-t rung beating the ensemble (-1.973 vs -1.933), reversing the NLL premium the chain paid on MNIST.

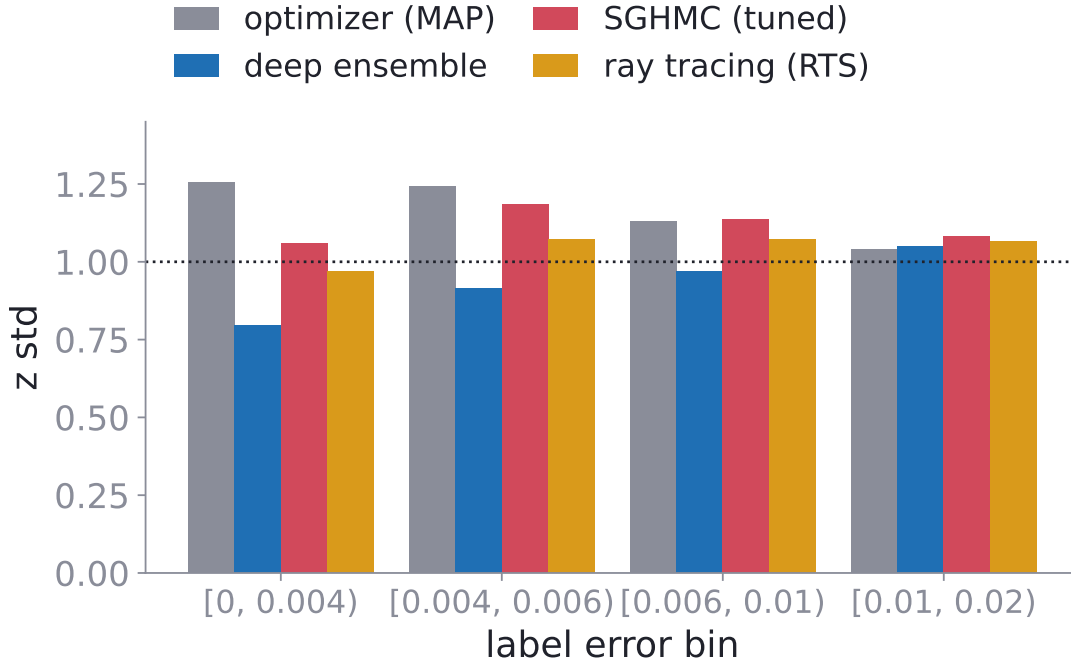


Figure 2 Per-bin calibration on the identical heteroscedastic likelihood. A calibrated method sits at the dotted line in every label-error bin. MAP is an overconfident staircase, the ensemble an inverted one; only the ray tracing chain is flat.

3.5 SGHMC and the temperature story

SGHMC is the natural head-to-head: it is the standard way to run Hamiltonian dynamics on minibatch gradients, and its friction term must absorb exactly the gradient-noise heating that ray tracing’s constant-speed construction removes. The comparison was tuned honestly: a pilot grid found that low friction heats catastrophically (kinetic temperature rising to 9.6x the target across step sizes), while a 100x larger friction holds temperature near 1, and both production arms carry a per-step kinetic thermometer (Figure 4). The tuned survivor still runs 8 to 9% hot for its entire 2M steps, its weighted norm is still marching at the end (no convergence certificate at this budget), and its calibration is worse than the chain’s in every bin with 170 tail stars. The deliberately under-damped arm boils (final-quarter kinetic temperature 2.9, excursions past 100), equilibrates 14.5k nats above the posterior, and still posts a good RMSE of 0.0474: point accuracy is no evidence of a correct posterior. The claim this supports is deliberately narrow: SGHMC must tune a friction against the gradient-noise scale to hold temperature and is measurably hot even then, while ray tracing holds temperature by construction with no noise-matched knob. Fair caveats: the SGHMC budget (2M steps) is smaller than the chain’s full campaign, and this is one target. On mixing we report the slowest observable honestly: integrated autocorrelation times are of order 1.2 to 1.4M steps for the chain (window-limited) and beyond 2M for SGHMC (never equilibrated); the fast observables both samplers like to advertise are deceptive by design.

3.6 The Student-t rung fixes the tail shape

The flat Gaussian chain still carried 108 held-out stars past $|z| = 4$ (chance: 1.6) with z kurtosis 8.1, and $\sigma(x)$ had already absorbed the ones it could: what remained was a shape problem, not a scale problem. The final rung samples a Student-t likelihood with scale $\sigma(x)$ and degrees of freedom ν as a parameter. The chain converged at 8.5M steps with $\nu = 5.48$ (16/84 interval 5.43 to

Table 2 All methods on the held-out stars. MAP, ensemble, and SGHMC fit the identical per-star likelihood with the identical warm-start schedule. Chain rows use the mixture predictive. The Student-t tail count uses the fitted t's own tail probability (chance 1.6); its bracketed z std is the variance-based analog.

method	RMSE	test NLL	SD(z)	z kurtosis	tails
Adam MAP	0.0489	-1.733	1.18	15.9	209
deep ensemble (10)	0.0473	-1.933	0.96	6.6	61
SGHMC (tuned)	0.0475	-1.875	1.16	10.3	170
ray tracing, Gaussian	0.0494	-1.941	1.08	8.1	108
ray tracing, Student-t	0.0480	-1.973	(1.12)		14

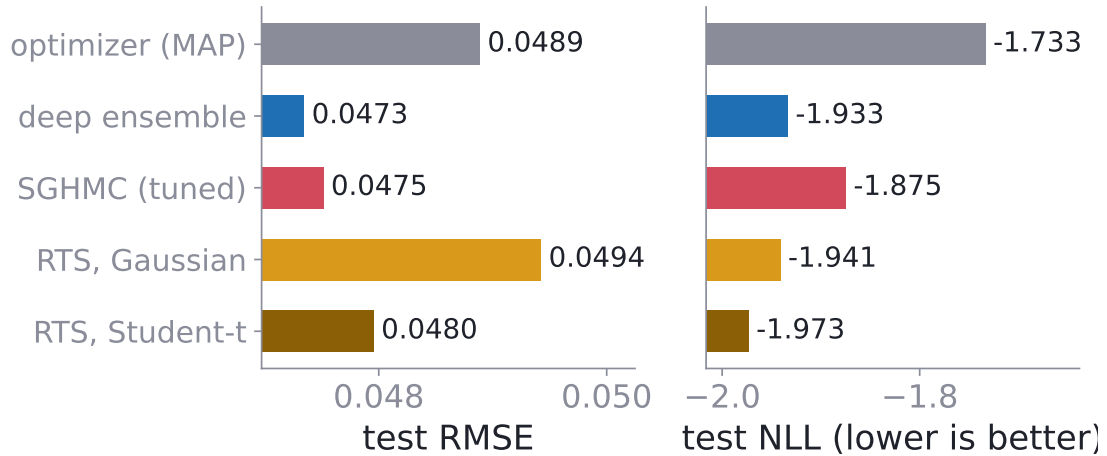


Figure 3 The predictive scoreboard. The ensemble keeps the best point accuracy; the two ray tracing chains post the two best test log-likelihoods, the metric that prices uncertainty.

5.53, close to the kurtosis-implied 5.1), improved RMSE to 0.0480 (outliers stop bullying the fit), and cut the tail count from 108 to 14 under its own tail probability (Figure 5). Width honesty did not change (roughly 4 to 5% under-coverage, shared with the Gaussian chain): the shape item is closed, the width item is the open one.

3.7 Sanity checks: alarms and physics

Two checks close the loop. First, out-of-distribution alarms: the chain's member-disagreement alarm scores AUC 0.93 on dwarf-star imposters and 1.00 on hot stars, and near chance (0.59) on survey-flagged stars, which is a label-side problem no spectrum-based alarm can see. The same panel is a fairness result: the ensemble and SGHMC alarms are statistically indistinguishable from the chain's, while the MAP fit's learned $\sigma(x)$ fails as an alarm entirely (it flags 8% of dwarfs, below the base rate), so out-of-distribution detection separates methods with genuine member spread from methods without, rather than ranking samplers; what the exactness certificates add is the guarantee that the chain's spread is a posterior, not an artifact. Second, physics: at fixed temperature and metallicity, high-alpha stars are 20 to 30% brighter below 450 nm in shape-normalized spectral stacks, and the network's saliency, read through the calibration (which is linear in the coefficients), concentrates in the same blue region. The model is looking where the physics says the signal lives.

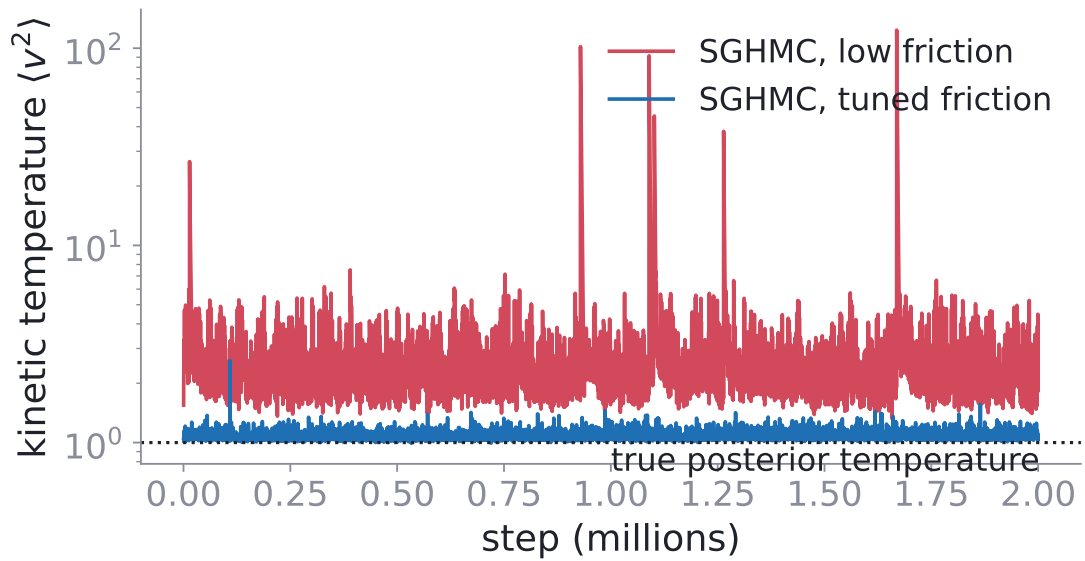


Figure 4 The kinetic thermometer $\langle v^2 \rangle$ for both SGHMC arms over 2M steps. The under-damped arm boils; the tuned arm sits just above the true posterior temperature (dotted) for the whole run, 8 to 9% hot. Ray tracing has no analog of this plot because its speed is constant by construction.

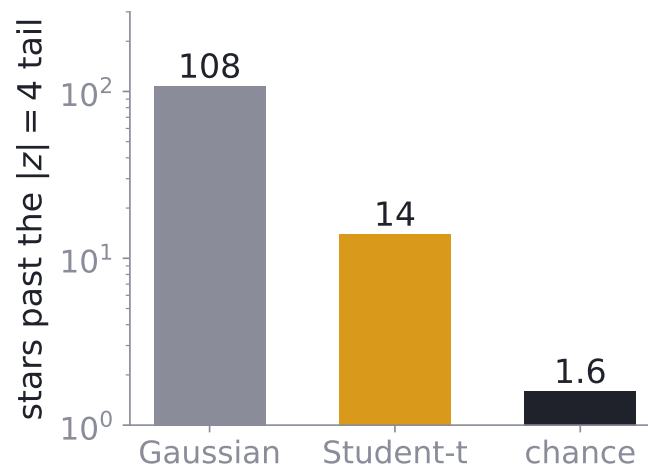


Figure 5 Held-out stars past the $|z| = 4$ tail under each likelihood's own yardstick. The sampled- v Student-t cuts the Gaussian chain's 108 to 14, against a chance expectation of 1.6.

4 Discussion

Three lessons carry the fortnight. First, calibration is a joint property of the likelihood and the sampler, and the two failures look different. The wrong noise model produces overconfidence no sampler can fix (the factor-of-seven ladder), while the right noise model in the wrong fitter produces structured miscalibration: staircases across error bins, inflated tails, and, most tellingly, an optimizer whose calibration worsens as its loss improves. The ladder fixes the aggregate; only sampling fixed the structure.

Second, the temperature story is the paper’s core claim made measurable. SGHMC has an explicit thermometer, and it reads hot: 8 to 9% in the best hand-tuned configuration, an order of magnitude in the under-damped one, with the friction knob needing to be re-tuned against whatever the gradient-noise scale happens to be. Ray tracing simply has no such knob to mis-set, and its converged chain passed an exact full-batch test at the end. On this target the practical benefit was not raw speed, it was that correctness did not have to be tuned for.

Third, evaluation at this grain is itself a contribution. Against the reference protocol of the Bayesian deep learning literature [6], this campaign reports the same predictive currency (RMSE and test log-likelihood) and out-of-distribution currency (AUC), exceeds the usual calibration granularity by treating the likelihood as an experimental variable and scoring per-group structure, and adds certificates the protocol lacks: the exact finisher, the gate’s measured noise ceiling, the kinetic thermometer, and drift rules that are only ever read next to acceptance. The honest converse is listed under limitations.

5 Limitations and next steps

Everything here is single-chain, so we have no multi-chain $\widehat{R}$, the central diagnostic of [6]; replicate chains from dispersed starts are cheap at this scale (hours, a few dollars each) and are the natural next run. The model is small ($D \approx 11,400$) next to the millions-of-parameters targets the literature now asks for; scaling the harness is a separate campaign. Posterior spreads remain lower bounds twice over: the slowest observables’ autocorrelation times are window-limited, and the gate runs at its measured noise ceiling, whose residual bias is one candidate for the one item the Student-t rung left open, the shared 4 to 5% width tightness (the other candidate is a richer $\sigma(x)$). The SGHMC comparison is one target at a smaller budget than the chain’s full campaign, and the survey-flagged imposter class shows every spectrum-based alarm is blind to label-side problems. Next steps: replicate chains for $\widehat{R}$, close the width item, and the standing direction from Report 2, auto-tuning from the misfit and norm signals, now with the noise ceiling as a third signal.

6 Conclusion

The harness left the toy. On a 126,156-star Gaia regression with real per-star errors, the minibatch ray tracing chain converged, passed an exact full-batch test, and delivered the only per-star-flat calibration among four ways of fitting the identical likelihood, with the Student-t rung closing the tail shape (108 to 14 outlier stars) and posting the best test log-likelihood in the study. The overconfidence that dominated the raw target was a modeling problem the likelihood ladder fixed; the structure was a sampling property the baselines could not buy back; and the temperature that SGHMC must tune for, and still misses by 8 to 9%, is held by construction here. The full campaign, tuning grids and dead ends included, cost about ninety dollars of cloud GPU time, and every number in this report is reproducible from the committed packs and seeds in the repository.

References

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